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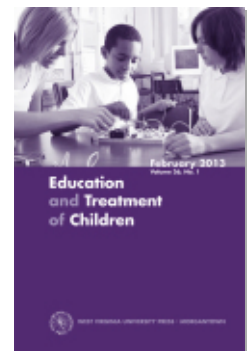
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Increasing Recreational Initiations for Children Who Have ASD Using Video Self Modeling

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Abstract

The current study examined the effects of video self modeling (VSM) on social initiations using a multiple baseline across participants experimental design. Three students with Autism Spectrum Disorders (ASD) watched videos of themselves engaged in typical, age appropriate social initiations prior to playtime with a triad of peers. A partial interval recording system was used to measure social initiations of students with ASD and typically developing peers. The percentage of intervals during which students with ASD exhibited play initiations increased to levels comparable to typically developing peers for all three participants. Two students with ASD demonstrated elevated levels of initiations during a maintenance probe conducted two weeks post-intervention. The efficiency of using VSM within a social skills curriculum, limitations of the current study, and future directions for VSM are discussed.

Autism is a complex neurological disorder diagnosed when an individual exhibits significant impairment in three broad areas of functioning: Social interactions, communication, and behavior (American Psychiatric Association, 2000). From Kanner's (1943) original conceptualization to the most recent Diagnostic and Statistical Manual of Mental Disorders (DSM-IV-TR), problems in social relatedness have been a characteristic of the disorder. Individuals diagnosed with ASD, have difficulty initiating interactions (Hauck, Fein, Waterhouse, & Feinstein, 1995), using appropriate greetings (Hobson & Lee, 1998), and establishing joint attention (McArthur & Adamson, 1996). These deficits underlie all social-behavioral adjustments across academic, personal, vocational, and community contexts (Gresham, 1982). Moreover, individuals with ASD may avoid or escape social interactions with peers, decreasing their opportunities to practice appropriate social interactions. Deficits in social relatedness persist across the lifespan (Njardvik, Matson, & Cherry, 1999) as the individual matures, the marked deviations from normal social interactions may continually increase. The cumulative impact of social deficiencies across the lifespan creates a need for effective evidence-based best

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practices that can be implemented early in a child's development to diminish barriers to the acquisition and fluent utilization of social engagements.

When children enter school settings, they are required to make two essential social-behavioral adjustments that involve compliance with the behavioral demands of classroom settings that are controlled by their teachers and adherence to the behavioral requirements and social dynamics of free-play settings that are controlled by peers (Williams, Walker, Holmes, Todis, & Fabre, 1989). Children with disabilities often do not have the necessary social skills to meet the minimal behavioral demands and expectations of less restrictive educational settings, nor are they able to initiate and maintain positive social relationships with key social agents within these settings (Gresham, 1982). In free-play settings, particularly on the playground, children and youth are required to display social skills in order to participate in ongoing peer-controlled activities and to gain peer acceptance. That is, to develop satisfactory peer relations and to fully participate, a student must be able to initiate interactions with peers, respond appropriately to others' initiations, assist others when the situation calls for it, interact with one or more peers for relatively long periods of time, know how to play games with minimal levels of competence, communicate effectively, and cope with peer provocations (Walker, Schwarz, Nippold, & Irvin, 1994). Children who are unable to display these behaviors are likely to be neglected and/or rejected by their peers and are at-risk for failure in their peer-related adjustments (Odom et al., 1992).

Appropriate social skills make it possible to achieve three important goals: allowing an individual to initiate and develop positive social relationships with others, facilitating the individual's ability to cope effectively with the behavioral demands and expectations of specific settings, and providing communication strategies to assert one's needs, desires, and preferences. Parallel movements of mainstreaming and deinstitutionalization of children and youth with disabilities have had a tremendous impact on the need for the direct teaching of social skills (Walker et al., 1994), yet, a missing component of many early intervention programs for children with ASD is effective, data-driven social skills training to address their specific social skills deficits (Hume, Bellini, & Pratt, 2005).

Video Modeling

Video modeling is often used in the acquisition and generalization of appropriate social and academic behavioral repertoires (Ayres,

& Langone, 2005; Bellini & Akullian, 2007; Charlop-Christy, Le, & Freeman, 2000; Nikopoulos & Keenan, 2003, 2004; Shukla-Mehta, Miller, & Callahan, 2010). Video modeling has been shown as an effective tool to teach children and adolescents with autism conversational speech (Charlop & Milstein, 1989), socially expressive behavior (Charlop, Dennis, Carpenter, & Greenberg, 2010), perspective taking (Charlop-Christy & Daneshvar, 2003), social initiations (Buggey, 2005; Nikopoulos & Keenan, 2003, 2004), as well as play skills (Nikopoulos & Keenan, 2003, 2004). Video production is done with confederates (e.g., peers or adults) demonstrating the target behavior(s). After completion the videos are shown to the participant in a training environment such as a classroom or home prior to observations. Results from video modeling interventions have been promising in the acquisition, transfer, and maintenance of target behaviors (Bellini & Akullian, 2007). However, learner and demonstrator characteristics can have a significant impact on the efficacy of generic video modeling.

Video Self-Modeling

Video self-modeling (VSM) has been defined as, “a procedure in which people see themselves on videotapes showing only adaptive behaviors,” (Dowrick, 1986, p. 201). VSM is a promising intervention that includes many aspects of Bandura’s social learning theory (Bandura, 1997, 2001; Charlop-Christy et al., 2000). In particular, VSM allows the individual to imitate targeted behaviors by observing her or himself successfully performing a behavior (Dowrick, 1999). Videotaping individuals acting as their own models was proposed as a method that maximizes characteristics that serve to promote attention to the model (Miklich & Creer, 1974). Individuals close to the observer’s age who have similar characteristics (gender, personality, race, and mood) and are functioning only slightly above the level of the observer have proven to be the most effective models (Bandura, 1997, 2001). Based on this research, VSM was developed to allow participants to view themselves in situations where they are performing at a more advanced level than they typically function.

The current study was designed to evaluate the effectiveness of VSM, and through replication, add to the growing literature support of VSM as an evidence-based practice for children with ASD (Shukla-Mehta et al., 2010). Intervals with initiations were compared within and across students with disabilities; initiation levels were also contrasted to typically developing peers within a playgroup. Persistence of initiations following two weeks without watching a VSM video was assessed via a maintenance probe.

Method

Participants

Three participants, 4–7 years old, participated in the study. Each participant was diagnosed within autism by an outside agency. Each participant's teacher reported relatively similar behavioral deficits, including: (a) limited social skill repertoires and (b) low frequencies of spontaneous social initiations.

Setting

All activities took place in the children's school. Participants observed the videos in a conference room adjacent to their classroom. The 12' by 13'8" conference room contained a television, a table in the center of the room, a chair for the participant, and a chair for the experimenter. Investigators videotaped assessment and intervention sessions in the school's play area through a one-way mirrored window. The 12' by 28' playroom contained several foam play mats, a 5' by 5' ball pen, and manipulative play materials.

Response Measurement

Social initiations were operationally defined as: (a) verbal statements by the participant directed toward peers in the playroom with the participant's head facing the peer or (b) independent verbal statements accompanied by the manipulation of a stimulus (e.g., asking a peer to play in the ball pit while holding a ball). A trained observer watched the videotapes and recorded the number of intervals containing social initiations by the participant using a pen and paper partial interval-recording sheet. The duration of each interval was 15 s and the duration of each video was 10 min. This duration for the intervals was chosen because the duration of an initiation ranged from three to eight s and partial interval recording scores each interval where an initiation occurred. So not to count repeat initiations of the same topography an inter-response time of five s per initiation was required. So the initiation duration and inter-response time between initiations ranged from 8 to 10 s, and allowing 15 s intervals for data collection gave a more accurate account of the number of independent initiations during a session. Observers were trained on the response definitions and assessed until they reached a 90% agreement criterion prior to observation sessions. The percentage of intervals with initiations was obtained by dividing the number of intervals with initiations by the total number of intervals for a session.

Procedure

A “hidden support” method of video creation was used; the primary investigator verbally prompted the participant to instigate a play-based initiation with a peer. Initiation responses were selected by observing the participant during free operant baseline sessions and using play preferences displayed by the targeted child or peer group. Subsequently, all prompts were edited from the videotapes prior to viewing so that the child appeared to independently ask peers to play. Participants viewed their individualized video in the aforementioned conference room 10 min prior to the start of the recess. The conference room was utilized for VSM monitoring to minimize distractions commonly observed within the classroom. If the child was not attending to the video (e.g., faced away from the video screen) they were verbally prompted by the experimenter to continue watching. The overall durations of the videos were 6 min and 43 s for participant one, 7 min and 27 s for participant two, and 6 min and 3 s for participant three. The number of clips for participant one was six, eight for participant two, and nine for participant three. After the video session the child was praised for attending to the video and brought to the playroom, where he joined his peers. Each peer triad included a target individual diagnosed with ASD, a second peer with ASD, and one peer who did not have an ASD.

Observation sessions were conducted during the first 10 min of the children’s playtime. Maintenance probes were conducted two weeks after the final intervention sessions; students did not view the videos during the time from last intervention session to maintenance probes.

Inter-Observer Agreement

During thirty-one percent of the assessment sessions, a second trained observer independently watched the session videos and recorded data on response initiations to assess inter-observer agreement (IOA). All sessions were taped and the sessions used for the data on IOA were chosen at random across all phases of the study. The percentage of inter-observer agreements was calculated using interval agreement ($\text{total agreements} / \text{total number of intervals} \times 100$). IOA on the percentage of intervals with an initiation for Jerry ranged from 70–98% with a mean of 80% and a Kappa value of .98. IOA on the percentage of intervals with initiations for Bob ranged from 90–100% with a mean of 96% and a Kappa value of .99. IOA on the percentage of intervals with an initiation for Phil ranged from 83–100% with a mean of 90% and a Kappa value of .96.

Table 1
Social Validity Scores for Participant Representatives

Question	Jerry		Bob		Phil		Overall
	Teach	Parent	Teach	Parent	Teach	Parent	
Prior Social Skills	5	4	2	3	3	2	3.17
Post Social Skills	9	8	7	7	9	8	8.00
Overall Change	7	7	7	6	9	9	7.50
Ease of Intervention	8	10	8	9	8	10	8.83

Social Validity

Following intervention, teachers and parents completed a short social validity survey. The Results for the 10-point anchored Likert scale (i.e., 1-low to 10-high) are presented in Table 1. Each participant had two respondents (i.e., a teacher and a parent). Questions asked on the survey included: 1) How would you rate the student/child's social skills prior to the intervention [Prior Social Skills], 2) How would you rate the student/child's social skills following the intervention [Post Social Skills], 3) How would you rate the overall change in the student/child's social skill from the beginning to the end of the interventions [Overall Change], and 4) How would you rate the ease of the intervention [Ease of Intervention]. All parents and teachers indicated that the social skills of participants seemed to have improved after the intervention was implemented. The overall change was rated as moderately high; the ease of intervention was rated very favorably.

Results

Figure 1 shows the percentage of intervals with an initiation across sessions for each of the three participants. Marked increases in spontaneous social initiation across all participants were observed after the implementation of VSM.

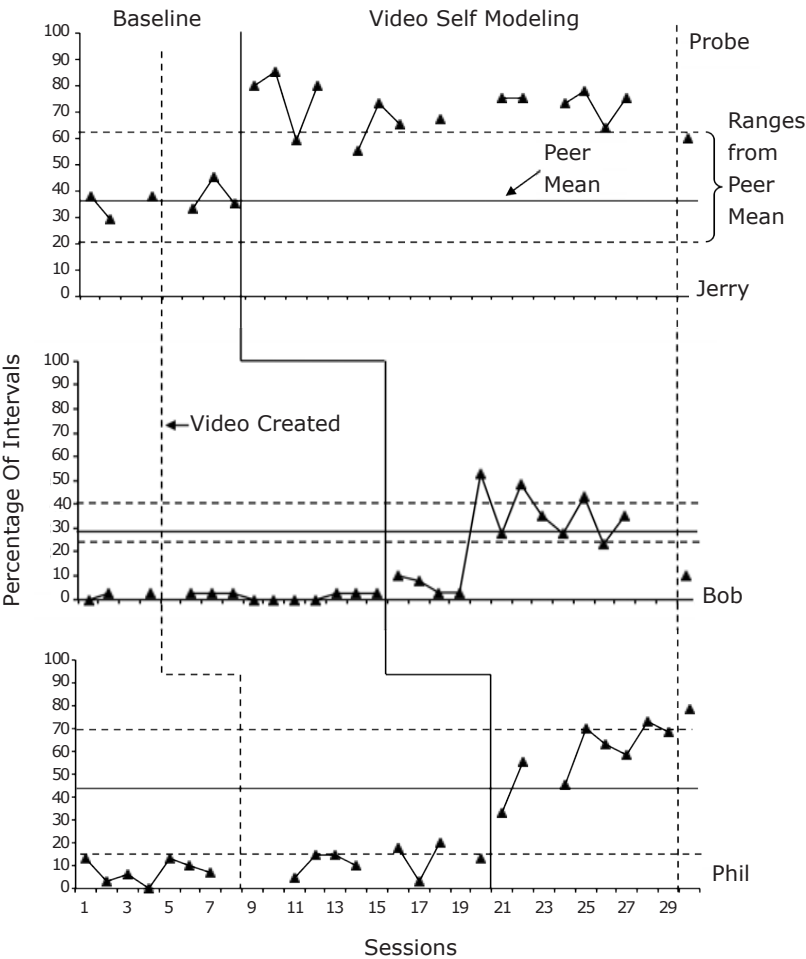


Figure 1. The percentage of intervals with initiations across participants. A horizontal solid line depicts the mean percentage for peers while the ranges of peer initiations are represented by dashed horizontal lines.

Jerry. During baseline, Jerry’s initiations were at a mean level of 36% of intervals. The data were stable around the mean with a slight increasing trend. After the introduction of the intervention, initiations rose to a mean of 72% of intervals. During intervention, the data remained stable with slight decreasing trend. During the maintenance probe, Jerry initiated play activities with his peers during 60% of intervals. While lower than the intervention mean (72%), it was 24% points higher than baseline.

Bob. Bob's baseline initiations were at a mean of 2% of intervals. The data were stable around the mean with no trend. During intervention, initiations rose from a mean of 2% to 26% of intervals. During the first four sessions of intervention, the intervention showed no effect on behavior. The fifth assessment showed an increase in initiation from 3% during the previous assessment to 53%. There was however a decreasing trend in the data from the fifth intervention session to the final intervention session. The maintenance probe showed a decrease in initiations to 10% of intervals that was well below the intervention mean but still five times greater than baseline levels. This decrease during maintenance may be an area for follow-up investigations.

Phil. Phil's baseline initiations were at 10% of intervals. The data were stable around the mean with no trend. During the intervention, initiations rose from a mean of 10% to 58% of intervals. The intervention data showed an increasing trend with the highest percentage of initiations, 78% of intervals, occurring during the maintenance probe.

Peer Comparison. For every participant, data on peer initiations were taken during 30% of the assessment periods. Data, obtained via the same paper and pen data collection instrument used for target participants, were collected from the recorded videos and sessions were chosen at random. The mean percentage of intervals for peers is reported on figure 1 by a horizontal solid line while dashed horizontal lines represent the ranges of peer initiations. Initiations for Jerry's peer dyad had a high of 61% of intervals and low of 21% of intervals with a mean of 36% of intervals. During baseline Jerry's mean level of initiation was equal to his peers' mean but rose above his peers' highest levels of initiation following the intervention. Initiation for Bob's peer dyad had a high of 40% of intervals and a low of 24% of intervals with a mean of 29% of intervals. During baseline Bob's mean level of initiation was under the low range of peers but increased to just below the peer mean during the intervention. Initiations for Phil's peer dyad had a high of 69% of intervals and a low of 15% of intervals with a mean of 42% of intervals. Phil's level of initiation was under the low range of peers but increased above the mean and rose to above the high range of peer initiation with the maintenance probe also being above the highest range of the peers.

Discussion

The present study gives further support to the benefits of VSM demonstrated by previous research (Ayres, & Langone, 2005; Bellini & Akullian, 2007; Buggley et al., 1999; Nikopoulos & Kennan, 2003, 2004; Shukla-Mehta et al., 2010). Editing videos was made simpler by the use of new technology (e.g., Moviemaker) further increasing

the efficacy of VSM. The participants appeared motivated to watch their self-modeling videos, which made the delivery of the intervention easy as reflected by the social validity scores. Watching the videos may induce a motivating operation increasing the effectiveness of social engagement with others as a contingency, thereby increasing the frequency of all behaviors that produce that engagement. The effects on behavior were immediate for two of the participants; increases in social initiations were seen in the third participant, Phil, within one week. The extent to which the intervention's effects on behavior will have an enduring change in the participants' behavioral repertoires is an empirical question that warrants further study. Initiations for Jerry and Phil increased to mean levels greater than their "typically developing" peers, further bolstering the power of this intervention as a behavior change procedure.

Limitations

There were several limitations of the study that should be noted. The videos that were created for each of the participants varied in length and the number of clips that were contained in each. Differences in the results seen could have been impacted by qualitative or quantitative differences in the videos. Properties such as the number of clips and the duration of the videos were not examined in the present investigation. However based on the author's anecdotal observations the durations of the videos could have been shorter. The participant's durations of attending to the videos decreased as the intervention sessions progressed accompanied by an increase in the frequency of non attending behaviors during the final clips of the videos. Research on these areas could increase the effectiveness and efficiency of the intervention.

Inter-observer agreement on Jerry's assessment videos ranged from 70–90%. He had a greater vocal repertoire than the other two participants and was very active which could have increased error in the raters' coding. A possibility exists that when teaching initiation behavior the participant may engage in the behavior at such a high rate that the response becomes aversive to peers. This was not directly assessed in the current investigation but warrants note as a precaution for future investigations.

Future Directions

Data on the attending behaviors to the videos were not taken. Although it appeared that the participants were watching the video screen, the extent to which they were attending to the video content could be questioned. It was only during the fifth intervention session

that Bob began to tact his behavior seen on the video (i.e., "That's Bobbie"). This evidence of his attending to the video coincided with clear increases in his initiations. The other two participants tacted their behavior as soon as they saw themselves during their first intervention video and their initiations increased immediately. While the introduction of the intervention for Phil may seem to coincide with Bob's improvement, the behaviors were independent (i.e., Bob and Phil did not attend the same classes nor play in the playroom at the same time). Bob's increase in initiations following his verbal acknowledgement of his behavior on screen taken together with similar increases in Jerry's and Phil's initiations when acknowledged seeing their own behavior seems to indicate that attending to the videos may be functionally related to changes in social behavior. Attending behaviors should be an area of future research in order to ensure treatment integrity for interventions that use videos. For example, integration of visual tracking methods could be used to quantify duration of "attending" behavior.

Summary

Effective, data-driven social skills training to target specific deficits, such as initiation, is often a component missing from many academic programs for children diagnosed with an autism spectrum disorder (Hume et al., 2005). The results gleaned from this examination of VSM provide further support for the efficacy of VSM for the acquisition of social initiations. A marked increase in the frequency of initiations was observed in all of the participants of the investigation enabling the child to make new friends and engage in typically developing activities with their peers. Future investigations may examine the ways in which a child with ASD attends to the VSM intervention in order to improve the efficiency of skill acquisition, the transition to fluency, and maintenance of social initiations.

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